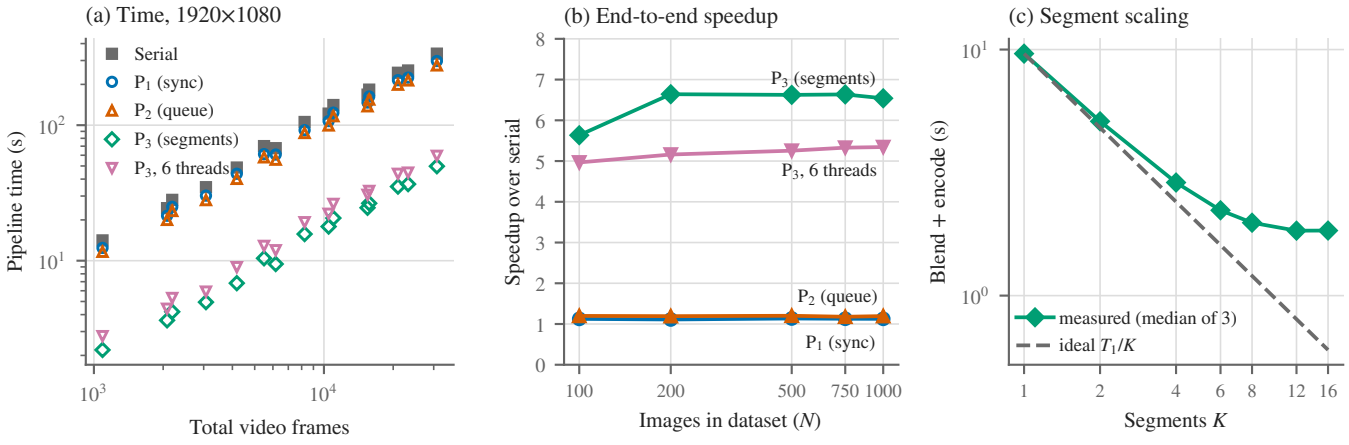
(a) Time, 1920×1080
Serial
P1 (sync)
P2 (queue)
P3 (segments)
P3, 6 threads
Pipeline time (s)
10^2
10^1
10^3
10^4
Total video frames
(b) End-to-end speedup
Speedup over serial
8
7
6
5
4
3
2
1
0
P3 (segments)
P3, 6 threads
P2 (queue)
P1 (sync)
100
200
500
750
1000
Images in dataset (N)
(c) Segment scaling
Blend + encode (s)
10^1
10^0
measured (median of 3)
ideal T1/K
1
2
4
6
8
12
16
Segments K